Customer Churn Prediction & Revenue Risk Prioritization

A Production-Oriented Machine Learning Pipeline for Retention Strategy

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1. Project Overview

Customer churn is one of the most critical drivers of revenue leakage in subscription and telecom-style businesses. This project develops an end-to-end machine learning solution that not only predicts which customers are likely to churn, but also quantifies the **financial impact of that churn** to help businesses prioritize retention efforts based on revenue risk rather than probability alone.

Unlike traditional churn models that stop at prediction, this solution bridges the gap between **data science and business decision-making** by translating model output into actionable retention targeting.

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2. Business Objective

The goal is to answer three key business questions:

1. **Who is likely to churn?**
2. **How much revenue is at risk if they churn?**
3. **Which customers should we prioritize to maximize ROI of retention campaigns?**

This enables retention teams to shift from reactive churn handling to **proactive, value-driven intervention**.

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3. Dataset Description

The dataset contains customer-level behavioral, service usage, and revenue attributes.

Category	Examples
Customer Information	Tenure, service usage patterns
Behavioral Metrics	Call activity, service adoption
Financial Data	Monthly revenue
Target Variable	churn (1 = churned, 0 = retained)

The customer field is treated as a unique identifier and excluded from model training.

4. Methodology

The workflow follows a **production-grade ML pipeline design**:

Step 1 — Data Preparation

- Remove identifier leakage (customer column excluded from modeling).
- Separate:
  - **Target Variable:** churn
  - **Features:** All behavioral and financial predictors.
- Automatically detect:
  - Numerical Features
  - Categorical Features

Step 2 — Robust Preprocessing Pipeline

A ColumnTransformer is used to ensure consistent preprocessing across training and inference.

Numerical Features

- Missing Value Imputation → Median Strategy
- Feature Scaling → Standardization (improves model convergence)

Categorical Features

- Missing Value Imputation → Most Frequent Value

This architecture prevents data leakage and ensures reproducibility.

Step 3 — Model Selection

A **Logistic Regression classifier** was selected due to:

- Interpretability for business stakeholders
- Stability on tabular datasets
- Probabilistic outputs required for risk scoring
- Fast deployment suitability

Step 4 — Train/Test Strategy

A **Stratified Split (75/25)** is applied to preserve churn distribution:

train_test_split(..., stratify=y)

This ensures evaluation reflects real-world class imbalance.

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5. Model Evaluation

The model is evaluated using:

- ROC-AUC Score → Measures ranking ability
- Precision / Recall → Evaluates churn detection quality
- Classification Report → Operational performance insight

This combination ensures both statistical validity and business relevance.

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6. Business Risk Scoring Framework

Predicting churn alone is insufficient.
We transform predictions into **financial intelligence**.

6.1 Revenue-Weighted Risk

$$Revenue\ Risk = Churn\ Probability \times Monthly\ Revenue$$

This identifies customers whose churn would cause the greatest immediate loss.

6.2 Customer Lifetime Value (CLV) Risk

We estimate future exposure:

$$CLV = Revenue \times Expected\ Remaining\ Months$$

$$CLV\ Risk = CLV \times Churn\ Probability$$

This quantifies long-term financial impact.

6.3 Final Retention Priority Score

A normalized hybrid score balances **likelihood** and **financial severity**:

$$Final\ Score = 0.6 \times Normalized(Revenue\ Risk) + 0.4 \times Churn\ Probability$$

This weighting emphasizes revenue protection while still considering churn likelihood.

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7. Customer Segmentation for Actionability

Customers are automatically categorized into business-friendly tiers:

Tier	Meaning
LOW	Monitor only
MEDIUM	Light engagement
HIGH	Targeted retention
CRITICAL	Immediate intervention

This segmentation allows marketing and CRM teams to operationalize results instantly.

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8. Revenue Exposure Analysis

The model calculates:

- Total Revenue Base
- Revenue At Risk
- % Revenue Exposure from High-Risk Customers

This provides leadership with **quantifiable justification for retention investment**.

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9. Deliverable Output

The system exports a ready-to-use operational file:

retention_target_list.csv

This file contains:

- Customer ID
- Churn Probability
- Revenue Contribution
- Financial Risk Metrics
- Priority Tier

It is designed for direct ingestion into CRM, campaign tools, or BI dashboards.

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10. Project Structure

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├── data/
│   └── cell2cell.csv
├── notebooks/
│   └── exploration.ipynb
├── src/
│   └── churn_pipeline.py
├── outputs/
│   └── retention_target_list.csv
├── requirements.txt
└── README.md
```

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11. How to Run the Project

Install Dependencies

pip install -r requirements.txt

Execute Pipeline

python churn_pipeline.py

Output Generated

retention_target_list.csv

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12. Key Strengths of This Solution

✓ End-to-End ML Pipeline (Not a notebook prototype)
✓ Business-Aligned Risk Modeling
✓ Revenue-Driven Decision Framework
✓ Fully Reproducible Preprocessing
✓ Deployment-Ready Architecture
✓ Bridges Data Science → Business Execution

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13. Assumptions

- Revenue is a reliable proxy for customer value.
- Expected remaining lifetime assumed as 18 months (can be tuned).
- Logistic Regression chosen for interpretability over complexity.

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14. Future Enhancements

- Gradient Boosting / XGBoost for higher predictive lift
- Survival Analysis for dynamic lifetime estimation
- Real-time scoring API deployment
- Integration with BI tools for executive dashboards
- A/B testing framework for retention campaign ROI validation

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15. Business Impact

This project transitions churn modeling from an academic exercise to a **profit-protection engine** by:

- Identifying high-risk customers before loss occurs
- Prioritizing retention budget efficiently
- Quantifying revenue exposure for leadership decisions

- Enabling scalable, data-driven customer management

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16. Author’s Note

This implementation reflects a practical understanding that machine learning in industry must deliver **actionable financial value**, not just predictive accuracy. The design intentionally emphasizes interpretability, scalability, and business alignment to support real-world deployment scenarios.
